

# **AI-Based Freelance-Startup System Progress Presentation**

Week 05

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# 01 Project Positioning

# Current Phase

Phase 1: Intelligent Foundation

Focus: System Design and Core Architecture

Goal of Week 04: Complete the implementation of this phase

## Phase Tasks:

AI Recommendation Tool (Candidate Match)

Profile Analyzer Tool

AI Predictive Tool

Selection and Mission Update

Mission Deployment

Real -Time notification system

Profile Engineering and Authentication

# Weekly Objective

01

## Complete the Objectives of Each Service

Ensure all microservices (Mission, Analytics, AI Orchestrator, etc.) fully implement their planned functionalities for Phase 1.

02

## Improve the Objectives of Each Tool

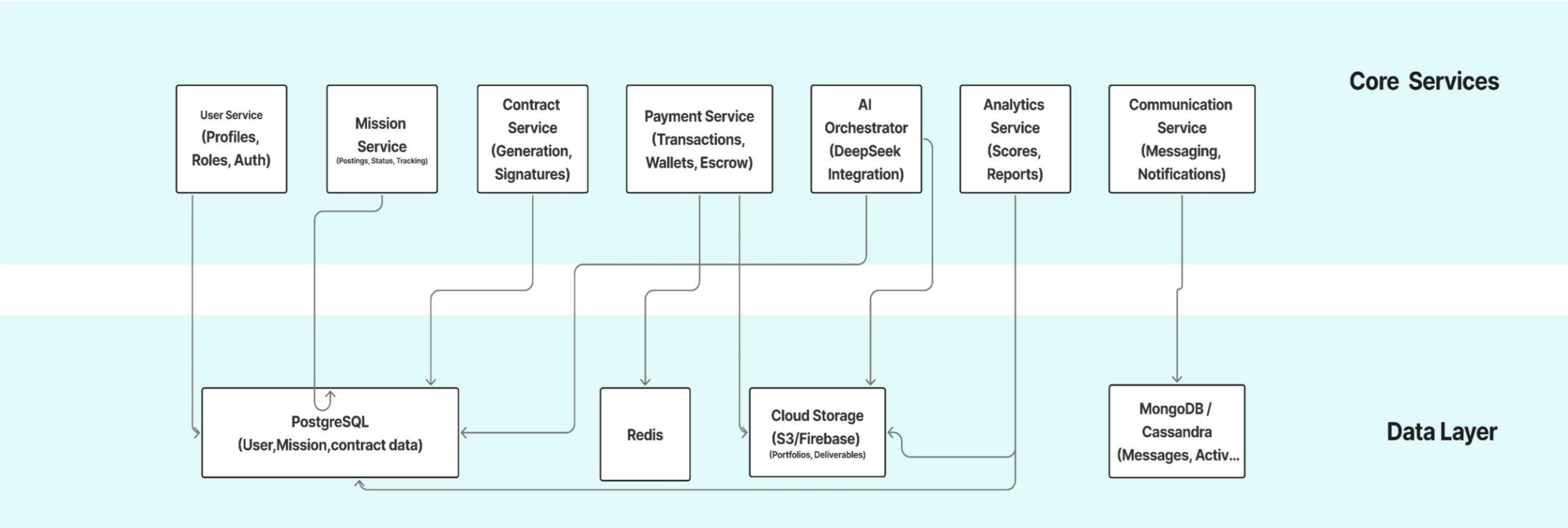
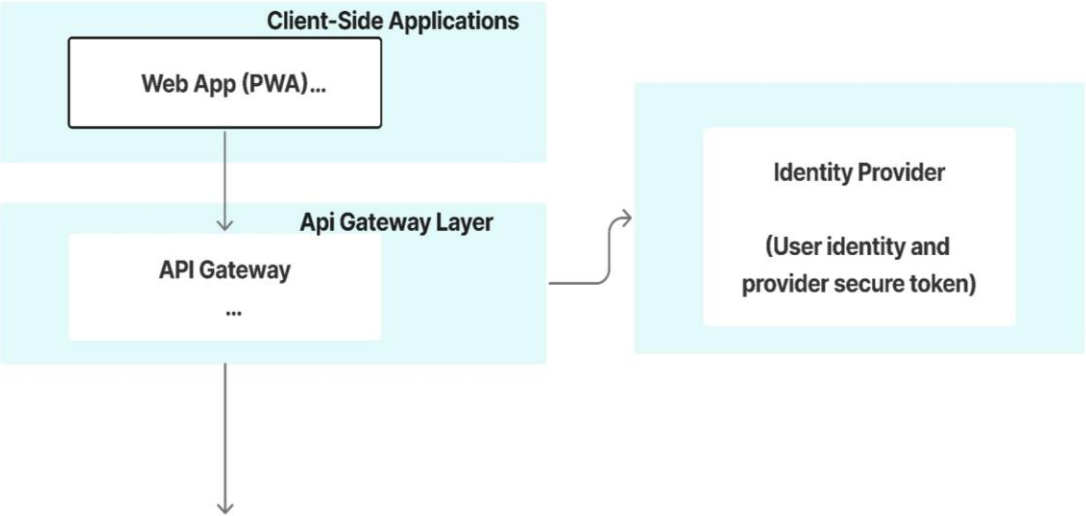
Improve and test tools (Profile Analyzer, AI Recommendation modules) to meet functional requirements.

02

## Integrate Frontend with Backend

Connect the frontend interface to the backend services for all features developed in this phase, enabling end-to-end functionality.

# Project Architecture







02

# Weekly Progress

## 2.1 Core Backend Services

### User Service

1

#### Authentication:

Developed Login and Sign up endpoints secured with JWT-based Authentication.

2

#### Startup Profile Engineering:

Implemented profile management endpoints of profile.  
Enable profile retrieval using authenticated user token.

3

#### Automatic Role initialization:

Designed a role initialization component to automatically seed roles in the PostgreSQL database at service startup .



## 2.1 Core Backend Services

### Mission Service

1

#### Mission Retrieval and Filtering:

Developed a backend API to fetch missions of company.

2

#### Auto-Termination Logic:

Implemented a timeout function to automatically closed missions that exceed their set duration.

3

#### Update Publish Mission:

Update publish mission feature to support secure access using JWT authentication.

4

#### Real -Time notification system:

Developed a Real-Time Notification System that instantly delivers Ai matching result to both companies and freelancers.

## 2.1 Core Backend Services

### Analytics Service

1

#### Reviews Management :

Submit, view, and report reviews between freelancers and startups after missions

2

#### Analytics Freelancer:

Full statistics and dashboard (ratings, success rate, performance score)  
Automatic recalculation after each new review.

3

#### AI Match Scores :

Process AI scores and return ranked freelancers per mission.

## 2.1 Core Backend Services

### Contract Service

1

#### Contract Management:

**Auto-Generation:** Automatically creates a contract when a mission is assigned to a freelancer, preventing duplicate contracts.

**Terms Generation:** Dynamically generates standard legal contract terms (scope, payment, IP, dispute resolution) based on mission data.

**Status Updates:** Allows updating contract status (Active, Completed, Cancelled) with optional reasons and timestamps.

## 2.1 Core Backend Services

### Contract Service

2

#### Contract Management:

**Auto-Milestone Creation:** Intelligently generates milestones based on mission duration ( 1 milestone for  $\leq 7$  days, 5 milestones for  $> 14$  days ..).

**Financial Allocation:** Calculates and assigns monetary amounts to each milestone based on configurable weights relative to the total budget.

**Timeline Mapping:** Automatically calculates and assigns due dates for milestones proportionally across the contract's duration.

**Milestone Status Tracking:** Provides methods to update individual milestone statuses (Completed, Approved) with completion timestamps.

## 2.2 AI and Intelligent Layer

### Integrated Tools

#### 1. Freelancer Tools

search freelancer  
get freelancer profile

#### 2. Company and Mission Tools

get company profile  
search mission  
get mission details

#### 3. AI Matching and Analysis

calculate compatibility score  
find best matching mission  
get matching missions

#### 4. Advanced Deep Analyzers

analyser for freelancer profile

## 2.3 Infrastructure and System Integration

### Containerization:

Dockerized Frontend, Gateway, User Service ,Mission Service and analysis service

### Gateway Routing:

linked User , Mission and Analysis Services to API Gateway

### RabbitMQ Messaging:

Asynchronous communication for workflow  
simulation(mission,analysis,Contract)

## 2.4 Frontend Layer

### Dashboard and Navigation

- Role-based dashboard (/dashboard)
- Freelancer view: applications and status overview
- Company view: missions and applications summary
- Sidebar navigation adapted per role

### Application Tracking: (Real Time)

- Application status tracking: (Pending, Viewed, Shortlisted, Accepted, Rejected)
- Dashboard with counts + detailed list
- Near real-time updates via WebSocket (STOMP + SockJS)



## 2.4 Frontend Layer

### Startup features

#### **Mission Creation**

- Multi-step “Post Mission” flow (/publish)
- Form validation before submission
- Deadline input with date and time (normalized format)

#### **Missions Management**

- Missions overview page (/missions)
- Status columns: Open, In Progress, Completed, Closed
- Search + quick navigation

### Freelancer features

#### **Browse and Apply**

- Browse missions (/applymission)
- Search functionality
- Apply with cover letter and proposed rate

#### **AI Suggestions**

- AI-recommended missions (/ai-suggestions)
- Match info (budget, deadline, score)
- Apply directly from suggestions



03

# Planning and Next steps

# Plan for the Next Week

- 1 Integrate all Phase 1 components into a fully functional system (main objective).
- 2 Launch Phase 2 and Phase 3 concurrently to accelerate overall development.

**Thank you**